Oscar Ramirez

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https://github.com/oscarramirezjr55/BAYESIAN-BACKPROPAGATION-Project

Education

- **California State Polytechnic University, Pomona (CPP)** — Pomona, California
  BACHELORS in Applied Mathematics/Statistics; GPA: 3.65 — *August. 2020 – May. 2024*
- **California State Polytechnic University, Pomona (CPP)** — Pomona, California
  MASTER'S in Applied Mathematics/Statistics; GPA: 3.8 — *August. 2024 – May. 2026*

Projects

- **Bayesian Backpropagation Deep Learning Model (Nov, 2025 – Dec, 2025)**: Developed a project to forecast 95% credible intervals for the standard error of missile interception with enemy drones using distance and material-based variables (concrete compressive strength, MPa). Implemented Bayesian neural networks using Python (NumPy, Pandas, Matplotlib, PyTorch), applying Bayesian inference, cross-validation, variational inference, KL-divergence, Gaussian reparameterization, gradient descent derivations, Adam optimizer, and Monte Carlo sampling. Code and analysis available at https://github.com/oscarramirezjr55/BAYESIAN-BACKPROPAGATION-Project
- **Financial Loan Risk Analyst Project (Nov, 2025 – Dec, 2025)**: Created multiple logistic regression models to classify customer loan repayment risk based on demographics. Applied statistical inference and model evaluation using Wald z-tests, K-fold cross-validation, AIC values, likelihood ratio tests, ROC curves, and Area Under the Curve (AUC) metrics. Full code and analysis: https://github.com/oscarramirezjr55/BANKINGLOAN-RISK-ANALYSIS
- **Investment in Residential Property Analysis (Nov, 2023)**: Conducted financial analysis of residential real estate investment opportunities using Microsoft Excel to identify properties with high return-on-investment (ROI) potential. Evaluated the feasibility of Accessory Dwelling Unit (ADU) additions by modeling cost estimates, rental income projections, and cash flow scenarios. Full report: https://livecsupomona-my.sharepoint.com/:x:
  /r/personal/olramirez_cpp_edu/Documents/139%20S%20Mainmone%20Ave%20San%20Dimas%2C%20CA%2091773%
  20Property%20Report.xlsx?d=w838e0b2b93fc4058b7451a17bdf8f257&csf=1&web=1&e=UstLXq.
- **Mathematical Modeling Project (Nov, 2024)**: Developed and analyzed delay differential equation models to study logistic population growth dynamics using MATLAB tools, including ODE45 and pplane8. Investigated the impact of key parameters on system behavior, incorporating complex numbers and Euler's equations. Report: https://acrobat.adobe.com/id/urn:aaid:sc:VA6C2:1625d586-195d-4459-923d-986527f895f3.

Interest and Classes for Spring 2026

- **Bayesian Time Series Models**: Kalman Filter, EFK, UFK, Particle Filter, Non-Linear Bayesian Estimation
- **STA 5320-01 Linear statistical Models**: Review of selected linear algebra and probability background material, including projection matrices and the multivariate normal distribution; matrix formulation of the multiple linear regression model, estimation, hypothesis testing, predictions, restrictions, and rank deficiency; one-way, two-way, and higher-way analysis of variance, random effects and variance components; illustration with statistical software and matrix computation software such as R or MATLAB.
- **STA 5300- 01 Random Processes**: Gaussian processes, covariance matrices, mean and covariance properties, Wiener processes and white noise, and second-order stationary processes. Analysis of queueing systems with equilibrium results for single and multiple server queues. Almost periodic processes. Frequency distribution of second-order stationary processes. Linear filtering. Renewal processes. Method of stages. Priority queues.
- **STA 4990 - 01 Special Topics for Upper Division Students**: Data munging, including loading, cleaning, reformatting, and error checking; multivariate exploratory data analysis; principles of prediction, including predictive accuracy, the bias-variance tradeoff, over-fitting, out-of-sample validation, and cross validation; predictive methods for both quantitative and binary variables, including linear and logistic regression, penalized regression, discriminant analysis, tree-based methods, support vector machines, and neural networks.

Skills

- **Proficient**R, Python, SQL, MicrosoftEXCEL